

SplitFed Speech Emotion Classification

Comparative Experiment Tables

Summary tables

Table 1: Train/test split statistics for the E0 cross-corpus setup.

Dataset	Split	Samples	Actors	Anger	Non-anger	Anger share
CREMA-D	Train	5890	72	1006	4884	17.08%
CREMA-D	Test	1552	19	265	1287	17.07%
RAVDESS	Train	1140	19	152	988	13.33%
RAVDESS	Test	300	5	40	260	13.33%
SAVEE	Train	360	3	45	315	12.50%
SAVEE	Test	120	1	15	105	12.50%

Table 2: Aggregated dataset-level statistics for the E0 setup.

Dataset	Samples	Actors	Anger	Non-anger	Mean dur. (s)	Std. dur. (s)
CREMA-D	7442	91	1271	6171	2.5429	0.5060
RAVDESS	1440	24	192	1248	3.7007	0.3367
SAVEE	480	4	60	420	3.8395	1.0780

Table 3: Pairwise Maximum Mean Discrepancy (MMD) across datasets without normalization.

Dataset	CREMA-D	RAVDESS	SAVEE
CREMA-D	-	0.381637	0.389779
RAVDESS	0.381637	-	0.332834
SAVEE	0.389779	0.332834	-

Table 4: Pairwise Maximum Mean Discrepancy (MMD) across datasets after normalization.

Dataset	CREMA-D	RAVDESS	SAVEE
CREMA-D	-	0.009802	0.018052
RAVDESS	0.009802	-	0.017652
SAVEE	0.018052	0.017652	-

Table 5: Main quality comparison across methods.

Method	Accuracy	Anger F1	Macro F1	Precision	Recall	PR-AUC
Centralized	0.8920	0.6864	0.8102	0.6655	0.7409	0.7772
FedAvg	0.8966	0.6598	0.7990	0.7101	0.6645	0.7757
SFLv1	0.8824	0.6594	0.7938	0.6601	0.6952	0.7442
SFLour	0.8727	0.6498	0.7854	0.6259	0.7313	0.7691

Table 6: Local corpus transfer matrix: anger F1 on cross-corpus evaluation.

Train model	CREMA-D test	RAVDESS test	SAVEE test
CREMA-D	0.6982	0.1395	0.5366
RAVDESS	0.1486	0.5517	0.2381
SAVEE	0.3449	0.2798	0.6400

Table 7: Resource usage across the main E1 methods.

Method	Wall time (s)	CPU user (s)	CPU sys (s)	Transport bytes	Avg. msg. bytes	Peak RSS (GB)	Peak CUDA alloc. (MB)
Centralized	484.34	358.09	81.25	0	0	2.16	149.4
FedAvg	539.58	2774.04	148.84	2103453750	7011512.5	2.14	149.4
SFLv1	767.44	1692.78	308.31	24554716930	257657.1	1.93	205.2
SFLour	787.29	1657.40	303.29	24554699080	258063.0	1.93	250.7

Table 8: SplitFed workload-policy quality at round 50 (seed 42). Metrics in the last four columns are macro-averaged across corpora. Results are from one run per policy; no uncertainty estimate is available.

Policy	CREMA-D F1	RAVDESS F1	SAVEE F1	Anger F1	Macro F1	UAR	PR-AUC
Full Local Epoch	0.6541	0.5517	0.8387	0.6815	0.8139	0.8148	0.7884
Fixed Maximum Steps	0.6513	0.6774	0.7317	0.6868	0.8163	0.8313	0.7809
Balanced Workload	0.6113	0.6087	0.7000	0.6400	0.7848	0.8201	0.7202

Table 9: SplitFed workload-policy cost at round 50. Batches/round are listed as CREMA-D/RAVDESS/SAVEE. The fixed budget of 143 is the RAVDESS (middle client) epoch length at batch size 8. GB and MB use decimal units.

Policy	Batches/round	Wall time (s)	CPU user (s)	CPU sys (s)	Transport (GB)	Messages	Peak RSS (GB)
Full Local Epoch	737/143/45	787.29	1657.40	303.29	24.555	95150	1.930
Fixed Maximum Steps	143/143/143	307.28	842.08	129.82	11.566	45550	1.918
Balanced Workload	64/64/45	184.84	358.55	60.87	4.865	19950	1.913